

Notes: Massa, Reuter, and Zitzewitz (JFE, 2010)
When Should Firms Share Credit with Employees? Evidence from
Anonymously Managed Mutual Funds

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1 Summary

Massa, Reuter, and Zitzewitz study why mutual fund families sometimes disclose individual fund-manager names and sometimes instead describe funds as co-managed or anonymously team-managed. The paper interprets naming a manager as a decision to share public credit for fund outcomes. Sharing credit can help the firm because named managers attract attention, create a fund-specific story, and may receive stronger incentives to perform. But credit sharing also makes successful managers more visible in the outside labor market and increases their future bargaining power.

The empirical setting is U.S. mutual funds from 1993 to 2004, a period in which anonymous management became much more common. The paper combines CRSP mutual fund data, Morningstar manager-name classifications, media mentions, fund flows, returns, fund holdings, IPO allocation data, evidence on late trading and stale-price arbitrage, and hedge-fund industry data. The main comparisons are across sole-managed, co-managed, and anonymously managed funds.

The main evidence supports a credit-sharing trade-off. Sole-managed funds receive more media attention and higher flows, especially among no-load funds sold directly to investors. Their average net returns are not significantly higher, but their return gaps are more favorable, and the evidence suggests that mutual fund families strategically allocate some benefits toward named-manager funds. On the cost side, departures of named managers reduce future flows, and the shift toward anonymous management is strongest where the hedge fund boom increased outside opportunities for successful managers.

Overall, the paper argues that fund families do not merely disclose manager names as neutral information. Rather, naming is a strategic organizational and marketing choice. The family must decide whether the benefits of creating a star manager outweigh the costs of giving that manager a portable reputation.

2 Big picture

- **Question.** When should firms publicly credit individual employees for project outcomes?
- **Setting.** U.S. mutual funds, where the fund family can name one manager, name multiple managers, or report anonymous team management.
- **Key organizational asset.** A fund's track record can be owned more by the family or more by a named individual manager.
- **Benefit of naming.** Named managers may generate investor attention, media coverage, stronger flows, and stronger effort incentives.
- **Cost of naming.** Successful named managers become more marketable, increasing their rent extraction and departure risk.
- **Main empirical pattern.** Anonymous management rises sharply between 1993 and 2004, especially for new funds.
- **Interpretation.** Mutual fund families strategically reduce credit sharing when the outside labor market for successful managers becomes more attractive.

3 Economic mechanism

The paper’s mechanism is a simple but powerful contracting idea. A fund family can either attach a track record to an individual manager or keep that track record closer to the organization. If the manager is named, investors and the financial media can connect fund outcomes to that person. This can be useful for marketing because it gives investors a plausible story about why the fund might outperform. It can also give the manager stronger career incentives because good performance becomes personally visible.

The same visibility creates a cost. Once a named manager performs well, outside employers can more easily identify and bid for the manager. Hedge funds are especially relevant because they expanded rapidly over the sample period and offered attractive outside options to successful mutual fund managers. The fund family then either pays more to retain the manager or risks losing flows after a departure.

Anonymous management reduces the portability of individual credit. It may weaken the marketing value of the fund and make performance less attributable to a specific person, but it also reduces the bargaining power of any individual manager. The central trade-off is therefore not just disclosure versus nondisclosure. It is a choice about how a firm allocates ownership of reputation between the organization and the employee.

4 Data and sample construction

The core sample combines mutual funds from the CRSP Survivorship-Bias Free Mutual Fund Database with Morningstar manager-name data. The paper excludes money market funds and classifies each fund as sole-managed, co-managed, or anonymously managed. Because manager-name fields can be recorded differently across data vendors and over time, the paper carefully compares Morningstar and CRSP classifications and validates them against SEC filings in the appendix.

The study then links management structure to several outcomes:

- media mentions in the *New York Times*, *Money*, *Kiplinger’s*, *SmartMoney*, and *Consumer Reports*;
- monthly net flows, calculated from changes in total net assets net of fund returns;
- net returns, CAPM alphas, Carhart four-factor alphas, expenses, and turnover;
- holdings-based measures, especially the return gap between actual fund returns and the returns implied by lagged disclosed holdings;
- IPO allocations and dilution from stale-price arbitrage or late trading;
- manager turnover and the subsequent flow response;
- hedge fund assets by investment category and location, used to proxy for changes in outside labor-market competition.

5 Empirical design

The paper uses a sequence of comparisons that map naturally to the benefit-cost framework.

1. **Document the trend in management disclosure.** Table 1 and Appendix Table A1 show the decline in sole management and the rise of anonymous management.
2. **Predict management structure.** Table 2 uses probit regressions to study whether fund characteristics, family characteristics, and fixed effects predict anonymous management or co-management.
3. **Estimate marketing benefits.** Table 3 studies media mentions, and Table 4 studies monthly net flows. These tests ask whether naming a manager attracts investor attention and demand.
4. **Study performance and strategic family support.** Tables 5 and 6 examine returns, return gaps, expenses, IPO allocations, and dilution from stale-price arbitrage or late trading.
5. **Study bargaining-power costs.** Table 7 asks whether flows fall when named managers depart, especially after strong past performance.
6. **Use labor-market shocks.** Tables 8 and 9 exploit variation in hedge-fund growth across asset classes and geography to test whether outside opportunities accelerate the move to anonymity.

A recurring identification device is the use of narrow fixed effects. For example, objective-by-month fixed effects compare funds within the same investment category at the same time; family-by-year fixed effects compare funds within the same fund family and year. These specifications help distinguish management-disclosure effects from broad time trends, family policies, or asset-class differences.

6 Main findings

6.1 Anonymous management rises sharply

The first fact is descriptive but important. Using Morningstar classifications, anonymous management rises from about 4% of funds in 1993 to about 18% in 2004. Among newly started funds, the increase is from about 3% to about 30%. The same broad trend appears in CRSP, although CRSP records more funds as anonymous and fewer as co-managed. The appendix shows why the paper relies primarily on Morningstar classifications.

This rise in anonymity is the phenomenon the paper seeks to explain. The trend is too large to treat as a minor reporting detail. It suggests a broad organizational shift in how mutual fund families allocated credit for fund performance.

6.2 Named managers receive more attention and flows

The marketing evidence is strong. Sole-managed funds are more likely to receive positive media mentions than anonymously managed funds, and in many specifications co-managed funds also receive more attention than anonymous funds. This is consistent with the idea that named managers make funds easier to profile and easier to market.

Fund-flow regressions show that anonymous funds receive lower net flows than otherwise similar named-manager funds. The effect is especially large for no-load funds, where direct marketing to investors is more important. The paper reports that sole-managed no-load funds receive roughly 3–4% of assets per year in additional net flows relative to anonymously managed funds, depending on controls.

6.3 Average returns are similar, but return gaps differ

The return evidence is more subtle. Net returns, CAPM alphas, and Carhart alphas do not differ significantly between sole-managed and anonymously managed funds. This means that the higher investor demand for named-manager funds is not simply explained by large, obvious differences in after-expense performance.

However, the holdings-based return gap is more favorable for sole-managed funds. The return gap compares actual pre-expense returns to the return on the fund’s last disclosed holdings. A higher return gap can reflect more profitable interim trading, better use of information, or favorable within-family treatment. The paper finds that sole-managed funds have return gaps around four basis points per month higher than anonymously managed funds.

6.4 Fund families appear to support named-manager funds

The paper interprets part of the return-gap difference as strategic cross-subsidization within fund families. Named-manager funds receive more favorable allocations of underpriced IPOs. Among international equity funds, named-manager funds also appear to suffer less return dilution from stale-price arbitrage or late trading. These patterns are consistent with fund families allocating scarce benefits toward funds where the marketing payoff from visible success is greater.

This evidence is important because it shows that naming managers is not merely a passive disclosure decision. It can be linked to how the family allocates resources and protects certain funds.

6.5 Named-manager departures are costly

The key cost of naming managers is visible in the manager-turnover tests. When named managers depart, net flows fall, especially for funds with strong prior performance. This is consistent with investors attaching part of the fund’s value to the named individual rather than only to the fund family.

This result helps explain why a family might prefer anonymous management even if naming raises flows while the manager stays. The family faces a dynamic problem: naming a successful manager today creates a portable asset that the manager may later take into the outside labor market.

6.6 The hedge fund boom helps explain the rise of anonymity

The final set of tests links the trend toward anonymity to hedge fund growth. Hedge funds expanded rapidly in the sample period, but their growth varied across asset classes and locations. The paper uses this variation in a difference-in-differences spirit: if hedge fund growth increases outside options

for mutual fund managers, then mutual fund families should shift more toward anonymity in the same asset classes and labor markets.

The evidence supports this prediction. The rise in anonymous management is stronger where hedge fund assets grow more. The paper estimates that hedge fund growth between 1994 and 2004 can explain a meaningful share—roughly 10% to 40%—of the rise in anonymous management.

7 Interpretation and contribution

The paper contributes to the mutual fund literature by showing that manager disclosure affects product differentiation, investor demand, and intra-family resource allocation. It also contributes to organizational economics by treating credit sharing as a strategic decision. Publicly crediting an employee can increase incentives and demand, but it can also make the employee harder to retain.

The paper’s logic applies beyond mutual funds. Firms often decide whether to attach success to individuals or teams: star analysts, portfolio managers, scientists, consultants, lawyers, CEOs, and product developers. The Massa, Reuter, and Zitzewitz mechanism suggests that firms may intentionally reduce individual visibility when outside labor-market competition is strong.

A useful way to summarize the paper is: *public credit is both a marketing asset and a bargaining liability*. The optimal amount of credit sharing depends on which force dominates.

8 How to read the tables

- **Table 1** documents the rise of anonymous management and the decline in sole management.
- **Table 2** studies the fund and family characteristics associated with anonymous or co-management.
- **Table 3** estimates how management structure predicts positive media mentions.
- **Table 4** estimates how management structure predicts monthly net flows.
- **Table 5** compares returns, expenses, turnover, return gaps, and number of holdings.
- **Table 6** examines stale-price-arbitrage dilution, late trading, and IPO allocations as potential sources of return-gap differences.
- **Table 7** shows how named-manager turnover affects flows.
- **Table 8** documents hedge fund and mutual fund assets by broad investment category.
- **Table 9** links hedge fund growth to the increased use of anonymous mutual fund management.
- **Appendix Tables A1–A2** compare CRSP, Morningstar, and SEC-filing classifications of management structure.

9 Extracted figures and tables from the paper

Table 1

The use of sole-management, co-management, and anonymous management by U.S. mutual funds, 1993–2004.

This table reports the percentage of all existing and newly started mutual funds that are classified as reporting one manager name (sole-management), reporting two or more manager names (co-management), or reporting no manager names (anonymously managed). Panel B corresponds to the full sample of mutual funds in the CRSP Survivorship-Bias Free Mutual Fund Database (except money market funds). Panel A corresponds to the subset of these funds for which we possess the manager name variable from Morningstar. In both panels, the unit of observation is the mutual fund rather than the mutual fund share class. We define the newly started funds in year t as those for which CRSP lists the year of inception as t . Since the Morningstar classification likely better reflects the information available to investors, we use the actual values reported by Morningstar in most of our analysis (see Table A1). However, to better highlight the rise of anonymously managed funds, the numbers and percentages in this table are adjusted for time-series changes in the rules that CRSP and Morningstar use to classify a mutual fund's management structure. For the purposes of this table only, in 1993–1996, when Morningstar identified any fund with more than two named managers as anonymously managed, we use the distribution of transitions in management type between 1996 and 1997 to impute management type in 1993–1996. From 1993 to 1999, CRSP's manager name variable occasionally reports a firm name rather than a manager name. In this table, we use the distribution of transitions from firm names to sole management, co-management, and anonymous management to adjust the aggregate CRSP statistics.

	Adjusted management classification				Adjusted management classification			
	All funds	Sole-managed	Co-managed	Anonymous	New funds	Sole-managed	Co-managed	Anonymous
Panel A: Management classification according to Morningstar								
1993	2,102	71.0%	25.1%	3.9%	325	73.2%	24.0%	2.7%
1994	2,572	69.7%	26.4%	3.9%	336	61.0%	32.2%	6.8%
1995	2,866	67.8%	27.7%	4.4%	197	64.5%	31.0%	4.5%
1996	3,094	62.5%	32.1%	5.4%	188	65.4%	28.9%	5.7%
1997	3,345	57.8%	34.6%	7.6%	312	51.9%	33.3%	14.7%
1998	3,827	52.3%	37.0%	10.7%	140	42.1%	41.4%	16.4%
1999	4,082	49.8%	38.2%	12.0%	51	66.7%	17.6%	15.7%
2000	4,300	47.2%	39.7%	13.1%	105	55.2%	21.9%	22.9%
2001	4,423	45.4%	38.2%	16.4%	76	40.8%	38.2%	21.1%
2002	4,388	43.6%	45.3%	11.1%	38	52.6%	31.6%	15.8%
2003	4,276	41.6%	42.0%	16.5%	136	46.3%	26.5%	27.2%
2004	4,937	41.1%	41.1%	18.3%	148	37.8%	31.8%	30.4%
Panel B: Management classification according to CRSP								
1993	2,783	79.2%	13.7%	7.1%	408	81.6%	13.7%	4.7%
1994	3,223	76.4%	16.6%	7.0%	406	71.6%	18.6%	9.7%
1995	3,555	74.3%	18.8%	6.9%	317	72.7%	21.3%	6.0%
1996	3,690	68.0%	24.3%	7.7%	261	71.9%	18.6%	9.5%
1997	4,266	62.3%	29.6%	8.1%	471	60.5%	31.5%	8.0%
1998	4,518	59.5%	29.6%	11.0%	233	50.2%	31.3%	18.5%
1999	4,590	53.7%	29.0%	17.3%	76	51.5%	32.1%	16.4%
2000	4,977	49.3%	28.2%	22.4%	135	52.6%	19.3%	28.1%
2001	5,102	46.3%	28.3%	25.4%	94	42.6%	33.0%	24.5%
2002	4,903	43.0%	27.8%	29.2%	60	43.7%	18.3%	38.0%
2003	5,580	40.5%	28.3%	31.2%	175	46.3%	27.4%	26.3%
2004	5,779	39.1%	29.9%	31.0%	331	28.7%	44.1%	27.2%

Table 1. Use of sole-management, co-management, and anonymous management by U.S. mutual funds, 1993–2004.

Table 2. (continued)

	Co-managed (t)			Anonymously managed (t)		
	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: Changes						
Dependent variable:	Co-managed (t)			Anonymously managed (t)		
Fixed effects?	Year	Objective · Year	Family · Year	Year	Objective · Year	Family · Year
Pseudo R2	0.0407	0.0634	0.2015	0.0368	0.0624	0.2450
Sample size	31,222	30,902	24,158	31,222	29,941	13,942
Panel B: Changes						
Dependent variable:	Co-managed (t)			Anonymously managed (t)		
Fixed effects?	Year	Objective · Year	Family · Year	Year	Objective · Year	Family · Year
Pseudo R2	0.0407	0.0634	0.2015	0.0368	0.0624	0.2450
Sample size	31,222	30,902	24,158	31,222	29,941	13,942

Table 2 continued. Levels and changes specifications for management structure.

Table 2

Determinants of anonymous management versus co-management, 1994–2004.

This table reports marginal effects estimated via probit. The sample consists of all mutual funds from the CRSP Survivorship-Bias Free Mutual Fund Database (except money market funds) for which we possess the Morningstar manager name variable. The unit of observation is fund i in January of year t . The dependent variable equals one if Morningstar leads us to classify fund i as co-managed (columns 1–3 and 7–9) or anonymously managed (columns 4–6 and 10–12) in year t . "Anonymously managed ($t-12$)" is a dummy variable that equals one if Morningstar lists fund i as being managed by unnamed managers in month $t-12$. "Co-managed ($t-12$)" is a dummy variable that equals one if Morningstar lists fund i as being managed by multiple named managers in month $t-12$. (Sole-managed funds are the omitted category.) "Fund new to Morningstar (t)" equals one if fund i was in the CRSP database in month $t-1$ but is new to Morningstar in year t . Fund characteristics come from CRSP: "Index Fund ($t-12$)" is a dummy variable that equals one if the CRSP fund name includes the word Index or an abbreviation thereof. "No load ($t-12$)" is a dummy variable that equals one if CRSP lists fund i as charging a sales commission. "Expense ratio ($t-12$)" and "12b-1 fee ($t-12$)" are fund i 's lagged expense ratio and 12b-1 fee. "Ln Fund TNA ($t-1$)" and "Ln Family TNA ($t-1$)" are the natural logarithm of dollars under management by fund i and by its family in month $t-1$. "Turnover ($t-12$)" is lagged portfolio turnover. "Fund age in years (t)" is the number of years between fund i 's inception (according to CRSP) and month t . "Net return ($t-12$ to $t-1$)" is defined as the natural logarithm of one plus the return of fund i between months $t-12$ and $t-1$. "Net flow ($t-12$ to $t-1$)" is defined as the natural logarithm of one plus the growth in fund i 's TNA between months $t-12$ and $t-1$ minus "Net return ($t-12$ to $t-1$)". It is the continuously compounded growth in assets minus the continuously compounded net return. "Std dev net return ($t-12$ to $t-1$)" is the standard deviation of fund i 's net return over the prior 12 months. Standard errors are clustered on mutual fund family. Columns 1, 4, 7, and 9 include a separate fixed effect for each year; columns 2, 5, 8, and 10 include a separate fixed effect for each investment objective each year; columns 3, 6, 9, and 12 include a separate fixed effect for each mutual fund family each year. The number of observations varies across specifications because observations are dropped when, for example, there is no variation in the dependent variable within a particular family in a particular year. Significance at the 10%, 5%, and 1% levels (in a two-sided test) is denoted by *, **, and ***.

Dependent variable:	Co-managed (t)			Anonymously managed (t)		
	(1)	(2)	(3)	(4)	(5)	(6)
Index fund ($t-12$)	-0.204*** (0.037)	-0.201*** (0.038)	-0.121*** (0.048)	0.072* (0.047)	0.055 (0.047)	0.262*** (0.115)
No load ($t-12$)	0.025 (0.024)	0.023 (0.024)	0.041 (0.034)	0.013 (0.019)	0.008 (0.019)	0.003 (0.025)
Expense ratio ($t-12$)	-0.000 (0.018)	-0.003 (0.021)	0.000 (0.023)	-0.007*** (0.019)	-0.007*** (0.022)	-0.030 (0.026)
12b-1 fee ($t-12$)	0.073 (0.073)	0.066 (0.070)	0.021 (0.067)	0.027 (0.034)	0.035 (0.036)	0.034 (0.056)
Ln Fund TNA ($t-1$)	0.016*** (0.006)	0.017*** (0.006)	0.016** (0.007)	-0.002 (0.003)	-0.003 (0.003)	-0.006 (0.005)
Ln Family TNA ($t-1$)	0.005 (0.014)	0.002 (0.013)	-0.008 (0.000)	-0.008 (0.004)	-0.009* (0.004)	-0.009* (0.000)
Turnover ($t-12$)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000*** (0.000)	0.000 (0.000)

Table 2, first page. Determinants of anonymous management versus co-management.

Table 3

Determinants of positive media mentions, by publication, 1997–2002.

Each column reports marginal effects from a probit regression predicting a positive media mention in a single publication or, in column 6, for a positive media mention in any of the five publications. Mentions come from the "Investing With" article in the New York Times, Money Magazine's Money 100 List, and articles in Kiplinger's Personal Finance, SmartMoney, and Consumer Reports. The sample consists of all mutual funds from the CRSP Survivorship-Bias Free Mutual Fund Database (except money market funds) between January 1997 and November 2002 for which we possess the Morningstar manager name variable. The unit of observation is fund i in month t . We include a separate fixed effect for each investment objective each month. Most independent variables are defined in the notes to Table 2. In addition, we control for Morningstar ratings (which range from one star to five stars) from December of the prior year. Since Morningstar ratings can vary across share classes, we set our "five star fund" control variable, for example, equal to the share of fund i 's TNA that receive a five star rating in the prior calendar year. "Family's print advertising to assets ratio ($t-12$ to $t-1$)" is defined as family i 's total print advertising expenditures between months $t-12$ and $t-1$ divided by family i 's average TNA during the same 12-month period. "Own-publication advertising ($t-12$ to $t-1$)" is defined as family i 's total advertising expenditure in publication between months $t-12$ and $t-1$. We exclude this variable when predicting media mentions in Consumer Reports (which does not accept advertising), and in column 6, when we predict a positive media mention in any of the five publications. The advertising data were acquired from Competitive Media Research and are described in Reuter and Zitzewitz (2006); they are measured in millions of dollars. Standard errors are clustered on mutual fund family. Significance at the 10%, 5%, and 1% levels (in a two-sided test) is denoted by *, **, and ***.

Publication:	Positive media mentions					
	New York Times (1)	Money Magazine (2)	Kiplinger's (3)	SmartMoney (4)	Consumer Reports (5)	Any Positive (6)
Objective · Month combinations with mentions	139	85	199	546	76	798
Observations in those combinations	34,735	9,737	40,261	80,822	11,309	110,133
Anonymously managed ($t-12$)	-0.118*** (0.020)	-0.552*** (0.204)	-0.067*** (0.022)	-0.137*** (0.039)	-0.079 (0.106)	-0.240*** (0.050)
Co-managed ($t-12$)	-0.078*** (0.022)	0.076 (0.194)	-0.052*** (0.024)	-0.037 (0.031)	-0.157** (0.098)	-0.111*** (0.046)
No load ($t-12$)	0.005 (0.025)	1.011*** (0.319)	0.109*** (0.065)	0.204*** (0.058)	0.700*** (0.240)	0.554*** (0.095)
Expense ratio ($t-12$)	0.023* (0.012)	0.183*** (0.078)	0.027*** (0.010)	0.056*** (0.016)	0.065 (0.063)	0.099*** (0.030)
12b-1 fee ($t-12$)	-0.069 (0.048)	-1.323*** (0.569)	-0.235*** (0.102)	-0.127 (0.135)	-3.142*** (0.862)	-0.658*** (0.197)
Ln Fund TNA ($t-1$)	0.043*** (0.011)	1.062*** (0.191)	0.094*** (0.018)	0.236*** (0.022)	0.267*** (0.088)	0.380*** (0.031)

Table 3, first page. Management structure and positive media mentions.

Table 3. (continued)									
Ln Family TNA (t-1)	-0.031*** (0.009)	-0.326*** (0.073)	-0.017*** (0.010)	-0.064*** (0.018)	-0.083*** (0.044)	-0.118*** (0.025)			
Turnover (t-12)	-0.000 (0.000)	-0.001 (0.001)	-0.000 (0.000)	-0.000* (0.000)	-0.001 (0.000)	-0.000** (0.000)			
Fund age (t)	-0.001 (0.001)	-0.011 (0.009)	0.001 (0.002)	0.000 (0.001)	0.003 (0.003)	-0.000 (0.002)			
Net flow (t-12 to t-1)	0.001*** (0.000)	0.004*** (0.002)	0.001*** (0.000)	0.002*** (0.000)	0.004*** (0.001)	0.003*** (0.000)			
Net return (t-12 to t-1)	0.007*** (0.001)	-0.002 (0.003)	0.003*** (0.001)	0.007*** (0.001)	0.002 (0.003)	0.010*** (0.001)			
Std dev net return (t-12 to t-1)	-0.003 (0.006)	-0.076 (0.050)	0.004 (0.004)	0.023** (0.008)	0.003 (0.025)	0.013 (0.010)			
Prior-year's Morningstar rating: 1 star	-0.219** (0.091)	-1.694** (0.657)	-0.248** (0.120)	-0.072 (0.092)	1.227** (0.442)	-0.260** (0.126)			
Prior-year's Morningstar rating: 2 stars	-0.030 (0.082)	-0.868*** (0.340)	-0.120** (0.052)	-0.079 (0.075)	1.442*** (0.429)	-0.131 (0.090)			
Prior-year's Morningstar rating: 3 stars	0.169*** (0.043)	0.177 (0.244)	0.018 (0.043)	0.125** (0.067)	2.511*** (0.497)	0.455*** (0.075)			
Prior-year's Morningstar rating: 4 stars	0.220*** (0.054)	0.610** (0.307)	0.147*** (0.047)	0.378*** (0.064)	2.740*** (0.641)	0.810*** (0.087)			
Family's print advertising dollars (t-12 to t-1)	0.002 (0.005)	-0.043** (0.022)	-0.005** (0.002)	-0.006* (0.003)	0.021*** (0.010)	0.007 (0.007)			
Family's nonprint advertising dollars (t-12 to t-1)	0.002 (0.004)	-0.041** (0.021)	-0.004*** (0.002)	-0.006 (0.004)	0.016 (0.013)	-0.005 (0.006)			
Own-publication advertising dollars (t-12 to t-1)	-0.049 (0.038)	0.453*** (0.123)	0.147*** (0.054)	0.275*** (0.183)					
Ho: Anonymous-Co-managed	0.034**	0.004**	0.282	0.034**	0.500	0.008**			
Objective - Month fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes			
Clustering	Family	Family	Family	Family	Family	Family			
Pseudo R2	0.182	0.342	0.298	0.260	0.360	0.290			
Observed probability (times 100)	0.043	0.919	1.341	1.474	1.270	1.000			
Predicted probability (times 100, at x-bar)	0.119	0.667	0.096	0.252	0.364	0.423			

Table 3 continued. Media-mention regressions and test statistics.

Table 5
Anonymous management, fund returns, and fund characteristics.
In this table, we estimate the determinants of monthly returns and fund characteristics in a panel regression with a separate fixed effect for each investment objective each month. We restrict the sample to actively managed, non-specialty domestic equity funds for which we can estimate risk-adjusted returns. In addition, we exclude funds by Putnam, Profunds, and Rydex. Panel A focuses on fund returns and fund characteristics. The sample consists of all non-specialty domestic equity funds from the CRSP Survivorship-Bias Free Mutual Fund Database (except money market funds) for which we possess the Morningstar manager name variable. Panel B focuses on performance measures and characteristics that require matched portfolio holdings. Since we only possess matched portfolio holdings through 2002, the sample in Panel B excludes 2003 and 2004. For the return measures, the unit of observation is fund i in month t . "Net return" is measured as continuously compounded percentage points per month. "Prior holdings" and "Return gap" are calculated as in Kacperczyk, Sialm, and Zheng (2008): prior holdings return is the return of the holdings from the most recent disclosure date and return gap is the difference between gross fund return (net return plus expenses) and the prior holdings return. When the dependent variable is the expense ratio, turnover, or number of stocks, the unit of observation is fund i in January of year t . Expense ratio and turnover are measured as percentage points per year. The number of stocks is the number of U.S. stocks disclosed in the fund's most recent N-300 filing. The independent variables are defined in the notes to Table 2. All independent variables except fund age are lagged. Specifications focused on monthly returns include S&P investment objective-by-month fixed effects. Columns 4, 5, and 10 include objective-by-year fixed effects. Column 9 also includes mutual fund family fixed effects. Standard errors cluster on year-month or year, as appropriate. Significance at the 10%, 5%, and 1% levels (in a two-sided test) is denoted by *, **, and ***.

Panel A: Analysis of fund returns, expenses, and turnover						
Dependent variable:	Net return monthly	CAPM alpha monthly	Carhart alpha monthly	Expense ratio annual	Turnover annual	
Sample frequency:	1994-2004	1994-2004	1994-2004	1994-2004	1994-2004	
Sample period:	(1)	(2)	(3)	(4)	(5)	
Anonymously managed (t-12)	-0.007 (0.030)	-0.034 (0.031)	-0.030 (0.026)	-0.170*** (0.044)	-12.064*** (4.661)	
Co-managed (t-12)	-0.008 (0.018)	-0.015 (0.018)	-0.024 (0.019)	-0.028 (0.040)	-6.292 (4.177)	
No load (t-12)	0.037* (0.021)	0.036* (0.021)	0.034* (0.018)	-0.348*** (0.062)	9.306* (5.156)	
Expense ratio (t-12)	-0.010 (0.052)	-0.056 (0.060)	-0.047 (0.053)		7.345 (7.189)	
12b-1 fee (t-12)	-0.004 (0.073)	-0.007 (0.073)	-0.018 (0.071)		1.127 (17.781)	
Ln Fund TNA (t-1)	-0.033*** (0.009)	-0.023** (0.009)	-0.013 (0.009)	-0.121*** (0.040)	-7.453*** (1.362)	
Ln Family TNA (t-1)	0.021** (0.008)	0.011 (0.009)	0.006 (0.007)	0.076*** (0.017)	3.717** (1.590)	
Turnover (t-12)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000** (0.000)		
Fund age (t)	0.000 (0.001)	-0.000 (0.001)	-0.001 (0.001)	0.005 (0.005)	0.112 (0.137)	
Net flow (t-12 to t-1)	-0.001 (0.001)	-0.001 (0.001)	0.000 (0.001)	0.001 (0.000)	-0.078** (0.037)	
Lagged return measure (t-12 to t-1)	0.021 (0.019)	0.016 (0.018)	0.015* (0.008)	-0.006 (0.004)	-0.130 (0.131)	
Std dev net return (t-12 to t-1)	-0.107 (0.116)	-0.089 (0.077)	-0.055 (0.057)	0.075*** (0.029)	5.295*** (1.032)	
Ho: Anonymous-Co-managed	0.383	0.542	0.305	0.000**	0.206	
Objective - Month fixed effects?	Yes	Yes	Yes	Yes	Yes	
Objective - Year fixed effects?	-	-	-	-	-	
Clustering	Month	Month	Month	Year	Year	
Sample size	126,898	126,898	126,898	10,489	10,489	
Panel B: Analysis of fund holdings						
Dependent variable:	Net return monthly	Prior holdings monthly	Return gap monthly	Return gap monthly	# Stocks	
Sample frequency:	1994-2002	1994-2002	1994-2002	1994-2002	1994-2002	
Sample period:	(6)	(7)	(8)	(9)	(10)	
Anonymously managed (t-12)	0.013 (0.034)	0.051* (0.030)	-0.035* (0.020)	-0.056** (0.028)	45.563* (23.620)	
Co-managed (t-12)	0.008 (0.023)	0.009 (0.023)	-0.001 (0.012)	-0.006 (0.016)	0.932 (8.775)	
Ho: Anonymous-Co-managed	0.882	0.219	0.063*	0.071**	0.056*	
Control variables from Panel A?	Yes	Yes	Yes	Yes	Yes	
Objective - Month fixed effects?	Yes	Yes	Yes	Yes	Yes	
Objective - Year fixed effects?	-	-	-	-	-	
Family - Month fixed effects?	-	-	-	-	-	
Clustering	Month	Month	Month	Month	Family	
Sample size	96,029	96,029	96,029	96,029	6,926	

Table 5. Anonymous management, fund returns, and fund characteristics.

Table 4. Determinants of monthly net flows.									
In this table, we estimate the determinants of monthly net flows in a panel regression with a separate fixed effect for each investment objective each month. The dependent variable is the natural logarithm of one plus the change in TNA between months t and $t+1$ minus the natural logarithm of one plus the fund's return between months t and $t+1$, which is the continuously compounded rate of growth in fund i 's assets minus its continuously compounded monthly return. We start with all mutual funds from the CRSP Survivorship-Bias Free Mutual Fund Database (except money market funds) for which we possess the Morningstar manager name variable, but then limit the sample to the 99.84% of fund-month observations with continuously compounded inflows between -100% and +100%. "Load (t-12)" is equal to one minus "No load (t-12)". The other independent variables are defined in the notes to Table 2. The sample period in columns 1-3 is 1994-2004. Columns 4-6 restrict the sample period to 1997-2002, when we possess data on both lagged Morningstar ratings and lagged media mentions. Columns 7-9 control for fund i 's prior-year's Morningstar ratings and the number of positive fund media mentions in NYT, Money, Kiplinger's, SmartMoney, and Consumer Reports between months $t-11$ and t . Standard errors are clustered on mutual fund family. Significance at the 10%, 5%, and 1% levels (in a two-sided test) is denoted by *, **, and ***.									
Sample period:	1994-2004			1997-2002			1997-2002		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Anonymously managed (t-12)	-0.167*** (0.073)	-0.238*** (0.108)	-0.269** (0.151)	-0.237** (0.096)	-0.401*** (0.135)	-0.407*** (0.189)	-0.194*** (0.095)	-0.351*** (0.135)	-0.330* (0.184)
Co-managed (t-12)	-0.027 (0.042)	-0.083 (0.061)	-0.134 (0.083)	-0.030 (0.051)	-0.093 (0.070)	-0.118 (0.095)	-0.016 (0.048)	-0.072 (0.067)	-0.081 (0.088)
Anonymously managed (t-12) - Load (t-12)	0.136 (0.133)	0.118 (0.187)			0.338 (0.188)	0.320 (0.249)		0.323* (0.189)	0.286 (0.242)
Co-managed (t-12) - Load (t-12)	0.097 (0.077)	0.161 (0.109)			0.115 (0.087)	0.171 (0.125)		0.103 (0.086)	0.131 (0.117)
Load (t-12)	0.010 (0.058)	-0.039 (0.062)	-0.096 (0.085)	0.106 (0.066)	0.021 (0.070)	-0.012 (0.098)	0.212*** (0.072)	0.134* (0.074)	0.152 (0.100)
Expense ratio (t-12)	0.043 (0.072)	0.041 (0.073)	0.135* (0.077)	-0.014 (0.036)	-0.018 (0.037)	0.050 (0.040)	-0.001 (0.035)	0.005 (0.035)	0.077** (0.038)
12b-1 fee (t-12)	-0.072 (0.138)	-0.074 (0.138)	-0.087 (0.155)	-0.046 (0.135)	-0.037 (0.134)	0.053 (0.215)	0.042 (0.137)	0.049 (0.135)	0.136 (0.200)
Ln Fund TNA (t-1)	-0.170*** (0.022)	-0.171*** (0.022)	-0.056* (0.023)	-0.167*** (0.024)	-0.168*** (0.024)	-0.239*** (0.036)	-0.239*** (0.027)	-0.155*** (0.027)	-0.155*** (0.037)
Ln Family TNA (t-1)	0.160*** (0.017)	0.107*** (0.017)	0.091*** (0.023)	0.092*** (0.018)	0.092*** (0.018)	0.077*** (0.026)	0.101*** (0.018)	0.086*** (0.018)	0.086*** (0.024)
Turnover (t-12)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Fund age (t)	-0.005*** (0.001)	-0.005*** (0.001)	-0.024*** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.021*** (0.003)	0.002 (0.001)	0.002 (0.001)	-0.008*** (0.002)
Net flows (t-12 to t-1)	0.030*** (0.002)	0.030*** (0.002)	0.030*** (0.002)	0.030*** (0.002)	0.030*** (0.002)	0.028*** (0.002)	0.028*** (0.002)	0.028*** (0.002)	0.028*** (0.002)
Net return (t-12 to t-1)	0.064*** (0.003)	0.064*** (0.003)	0.087*** (0.004)	0.058*** (0.003)	0.058*** (0.004)	0.079*** (0.003)	0.055*** (0.003)	0.055*** (0.003)	0.071*** (0.004)
Std dev net return (t-12 to t-1)	0.011 (0.016)	0.011 (0.016)	0.000 (0.019)	-0.000 (0.016)	-0.000 (0.016)	-0.000 (0.018)	-0.000 (0.016)	-0.000 (0.016)	-0.000 (0.018)
Ho: Anonymous-Co-managed (All funds)	0.046**			0.023**		0.054**			
Ho: Anonymous-Co-managed (No load)		0.111	0.337		0.009**	0.101		0.025**	0.154
Ho: Anonymous-Co-managed (Load)		0.202	0.165		0.589	0.425		0.677	0.585
Objective - Month fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control for prior-year Morningstar ratings?	No	No	No	No	No	No	No	No	No
Control for prior media mentions?	No	No	No	No	No	No	Yes	Yes	Yes
Clustering	Family	Family	Family	Family	Family	Family	Family	Family	Family
R2	0.1222	0.1222	0.0923	0.1199	0.1200	0.0912	0.1226	0.1227	0.1000
Sample size	379,621	379,621	379,621	225,727	225,727	225,727	225,727	225,727	225,727

Table 4. Determinants of monthly net flows.

Table 6
Anonymous management, return dilution due to stale price arbitrage and late trading, and IPO allocations.
This table explores two potential explanations for the lower return gaps of anonymously managed funds. In the first three specifications, we focus on dilution due to stale price arbitrage and late trading in international equity funds. Within the small sample of international equity fund-months for which Lipper and TrimTabs daily flow data are available, dilution is calculated as in Zitzewitz (2008). Estimation is via OLS, and includes S&P investment objective-by-month fixed effects, as well as the full set of control variables from Table 5; column 3 includes mutual fund family-by-month fixed effects. In the remaining four columns, we focus on IPO allocations to anonymously managed funds. Our sample is restricted to non-specialty domestic equity funds between 1994 and 2002, and the unit of observation is fund i in quarter t . Following Reuter (2006) and Gaspar, Massa, and Matos (2006), we construct proxies for IPO allocations from reported holdings of recent IPOs. We consider four (related) measures of the contribution of IPOs to fund performance. The dependent variable in column 4 is a dummy variable that equals one if fund i reported holding shares in any of the IPOs that occurred during the past quarter. We estimate column 4 via probit and report marginal effects. The dependent variable in column 5 is the ratio of the value of fund i 's holdings of recent IPOs to the fund's end-of-quarter TNA. Since this variable equals zero much of the time and cannot be negative, we estimate column 5 via Tobit. The dependent variable in column 6 is a dummy variable that equals one if fund i reported holdings suggest that it earned positive returns from underpricing during the past quarter. We estimate column 6 via probit and report marginal effects. Finally, the dependent variable in column 7 is the total underpricing that we estimate fund i earned over the prior quarter based on reported holdings at quarter end, divided by the fund's end-of-quarter TNA. Since this variable can be negative, zero, or positive, we estimate column 7 via OLS; however, we trim the top 1% of the positive and negative values. Standard errors are clustered on date. Significance at the 10%, 5%, and 1% levels (in a two-sided test) is denoted by *, **, and ***.

Sample:	International equity, 1998-2003			Non-specialized domestic equity, 1994-2002			
Dependent variable:	Net return	Dilution	Dilution	Any IPOs dummy?	IPOs as fraction of fund TNA	Hot IPOs dummy?	Underpricing as fraction of fund TNA
Sample frequency:	monthly	monthly	monthly	quarterly	quarterly	quarterly	quarterly
Estimation:	OLS	OLS	OLS	Probit	Tobit	Probit	OLS
Report:	coefficients	coefficients	coefficients	marginal effects	coefficients	marginal effects	coefficients
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Anonymously managed (t-12)	-0.095 (0.089)	-0.027*** (0.009)	-0.061** (0.024)	-0.018*** (0.006)	-8.137*** (2.309)	-0.026*** (0.008)	-45.146** (17.484)
Co-managed (t-12)	-0.110 (0.086)	-0.012* (0.007)	-0.028* (0.014)	-0.010*** (0.005)	-2.036 (1.532)	-0.017*** (0.006)	-13.853 (15.236)
Ho: Anonymous-Co-managed	0.850	0.117	0.050**	0.090*	0.010**	0.090*	0.020**
Control variables from Table 5?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Objective - Month fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Family - Month fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Month	Month	Month	Quarter	Quarter	Quarter	Quarter
Sample size	6,336	6,336	6,336	77,138	45,139	30,331	44,474

Table 6. Return dilution, late trading, and IPO allocations.

Table 7

Monthly net flows and manager turnover, 1994–2004.

In this table, we study the impact of manager turnover on monthly net flows. The sample, dependent variable, fixed effects, and (most) other independent variables are the same as in column 1 of Table 4. "Return rank" ranges from zero, when fund has the lowest net return within its investment objective between months $t-12$ and $t-1$, to one, when it has the highest net return. "Named-manager turnover dummy" equals one if one or more named managers departs the fund, or it switches from sole-managed or co-managed to anonymous management. Column 2 adds "Return rank squared" and "Net return squared" to the specification in column 1. Column 3 excludes "Net flows". The hypothesis tests are conducted assuming a return rank of 0.5. For example, "Ho: Anonymous-Sole-managed" corresponds to the test that the coefficient on "Anonymously managed ($t-12$)" plus 0.5 times the coefficient on "Return rank ($t-12$ to $t-1$)" is zero. "Anonymously managed ($t-12$)" equals zero. Standard errors are clustered on mutual fund family. Significance at the 10%, 5%, and 1% levels (in a two-sided test) is denoted by *, **, and ***.

	(1)	(2)	(3)
Anonymously managed ($t-12$)	0.055 (0.122)	0.036 (0.130)	0.051 (0.167)
Co-managed ($t-12$)	0.018 (0.065)	0.008 (0.066)	-0.002 (0.084)
Return rank ($t-12$ to $t-1$)	2.249*** (0.132)	0.920*** (0.240)	0.903*** (0.284)
Return rank squared ($t-12$ to $t-1$)	1.111*** (0.233)	-0.397* (0.197)	1.578*** (0.281)
Return rank ($t-12$ to $t-1$) × Anonymously managed ($t-12$)	-0.448** (0.218)	-0.397* (0.215)	-0.521** (0.257)
Return rank ($t-12$ to $t-1$) × Co-managed ($t-12$)	-0.078 (0.118)	-0.060 (0.118)	-0.068 (0.140)
Named-manager turnover dummy ($t-12$ to $t-1$)	0.171 (0.110)	0.134 (0.111)	-0.058 (0.123)
Return rank ($t-12$ to $t-1$) × Named-manager turnover dummy ($t-12$ to $t-1$)	-0.684*** (0.244)	-0.609** (0.247)	-0.543*** (0.260)
No load ($t-12$)	-0.049 (0.056)	-0.059 (0.056)	-0.037 (0.079)
Expense ratio ($t-12$)	0.054 (0.064)	0.037 (0.062)	0.123* (0.066)
12b-1 fee ($t-12$)	0.013 (0.129)	0.014 (0.129)	0.014 (0.185)
Ln Fund TNA ($t-1$)	-0.185*** (0.022)	-0.189*** (0.022)	-0.085*** (0.031)
Ln Family TNA ($t-1$)	0.100*** (0.016)	0.106*** (0.017)	0.100*** (0.022)
Turnover ($t-12$)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Fund age (t)	-0.004*** (0.001)	-0.004*** (0.001)	-0.022*** (0.002)
Net flows ($t-12$ to $t-1$)	0.029*** (0.002)	0.029*** (0.002)	0.044*** (0.004)
Net return ($t-12$ to $t-1$)	0.024*** (0.004)	0.029*** (0.003)	0.087*** (0.004)
Net return squared ($t-12$ to $t-1$)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
Std dev net return ($t-12$ to $t-1$)	-0.013 (0.016)	-0.104*** (0.016)	-0.087*** (0.019)
Ho: Anonymous-Sole-managed	0.018**	0.023**	0.031**
Ho: Anonymous-Co-managed	0.030**	0.041**	0.073*
Ho: Co-managed-Sole-managed	0.611	0.576	0.533
Objective-month fixed effects?	Yes	Yes	Yes
Clustering	Family	Family	Family
R ²	0.1090	0.1106	0.0853
Sample size	379,621	379,621	379,621

Table 7. Monthly net flows and manager turnover, 1994–2004.**Table 9**

The growth of hedge funds and the anonymous management of mutual funds.

This table reports marginal effects estimated via probit, with the goal of testing whether the use of anonymous management is correlated with growth in the hedge fund industry. The sample consists of all mutual funds from the CRSP Survivorship-Bias Free Mutual Fund Database (except money market funds) for which we possess the Morningstar manager name variable. The unit of observation is fund i in January of year t . The dependent variable equals one if Morningstar lists fund i as anonymously managed in the same calendar year. All independent variables except Fund age are measured during the prior calendar year. The first hedge fund-related variable is the natural logarithm of hedge fund assets under management in the same broad asset class (i.e., debt, domestic equity, or international equity) as fund i . The second hedge fund-related variable is the natural logarithm of hedge fund assets managed by firms in the same state as fund i . The third set of hedge fund variables are dummy variables indicating whether fund i is located in Boston or New York City (NYC) plus interactions with the natural logarithm of total hedge fund industry assets under management. All hedge fund assets are measured in billions of U.S. dollars. (All other independent variables are defined in the notes to Table 2.) While the set of fixed effects varies across specifications, all specifications include a full set of year fixed effects. The number of observations varies across specifications (and relative to specifications in Table 2) because observations are dropped when, for example, there is no variation in the dependent variable within a particular family in a particular year. In columns 1–2, standard errors are clustered on S&P investment objective as reported in CRSP. In columns 3–6, standard errors are clustered on state. Significance at the 10%, 5%, and 1% levels (in a two-sided test) is denoted by *, **, and ***.

Specification:	Levels	Changes	Levels	Changes	Levels	Changes
Sample period:	1994–2004	1994–2004	1996–2002	1996–2002	1996–2002	1996–2002
	(1)	(2)	(3)	(4)	(5)	(6)
Ln Hedge fund AUM in same broad asset class ($t-12$)	0.012** (0.006)	0.023*** (0.009)				
Ln Mutual fund TNA in same broad asset class ($t-12$)	-0.008 (0.010)	-0.039 (0.027)				
Ln Hedge fund AUM in same state ($t-12$)			0.009** (0.004)	0.007** (0.003)		
Ln Mutual fund TNA in same state ($t-12$)			0.005 (0.004)	0.000 (0.004)		
Boston HQ (t)					-0.335** (0.132)	-0.286*** (0.135)
Boston HQ × Ln Hedge fund industry AUM ($t-12$)					0.087** (0.040)	0.066** (0.025)
NYC HQ (t)					-0.306* (0.161)	-0.270** (0.157)
NYC HQ × Ln Hedge fund industry AUM ($t-12$)					0.051* (0.032)	0.040* (0.018)
Anonymously managed ($t-12$)	0.843*** (0.015)	0.607*** (0.022)			0.611*** (0.023)	
Co-managed ($t-12$)	0.017 (0.011)	0.009 (0.007)			0.009 (0.007)	
Fund new to Morningstar (t)	0.241*** (0.049)	0.093*** (0.044)	0.093*** (0.044)	0.092*** (0.043)	0.092*** (0.043)	0.092*** (0.043)
Index fund ($t-12$)	0.226*** (0.042)	0.094*** (0.020)	0.118** (0.075)	0.059* (0.041)	0.114* (0.075)	0.054* (0.039)
No load ($t-12$)	-0.014 (0.019)	-0.009 (0.019)	-0.016 (0.019)	-0.013 (0.020)	-0.013 (0.012)	-0.011 (0.012)
Expense ratio ($t-12$)	0.075*** (0.022)	0.041*** (0.011)	0.032* (0.017)	0.019** (0.010)	0.018* (0.017)	0.018* (0.010)
12b-1 fee ($t-12$)	0.067** (0.032)	0.043** (0.021)	0.002 (0.032)	-0.002 (0.017)	0.003 (0.031)	-0.002 (0.016)
Ln Fund TNA ($t-1$)	-0.008* (0.004)	-0.005** (0.003)	-0.008*** (0.003)	-0.004** (0.002)	-0.008** (0.003)	-0.004** (0.002)
Ln Family TNA ($t-1$)			-0.008*** (0.003)	-0.010** (0.004)	-0.005 (0.004)	-0.008** (0.004)
Turnover ($t-12$)	0.000** (0.000)	0.000*** (0.000)	0.000** (0.000)	0.000*** (0.000)	0.000** (0.000)	0.000*** (0.000)
Fund age (t)	-0.001*** (0.001)	-0.000 (0.000)	-0.001*** (0.000)	-0.000 (0.000)	-0.001*** (0.000)	-0.000 (0.000)
Net flow ($t-12$ to $t-1$)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Net return ($t-12$ to $t-1$)	-0.000** (0.000)	-0.000 (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Std dev net return ($t-12$ to $t-1$)	0.000 (0.002)	0.002 (0.002)	-0.000 (0.002)	0.001 (0.001)	0.000 (0.002)	0.001 (0.001)
Fixed effects	Family-Year	Family-Year	Objective-Year	Objective-Year	Objective-Year	Objective-Year
Clustering	S&P Objective	S&P Objective	Family	Family	State	State
Pseudo R ²	0.2767	0.5926	0.3437	0.5653	0.3413	0.5658
Sample size	11,922	11,922	13,795	13,795	13,795	13,795

Table 9. Hedge fund growth and anonymous mutual fund management.**Table 8**

Sizes of global hedge fund and U.S. mutual fund industries, 1994–2004.

Hedge fund asset totals by investment category are from the IASS database, as reported by Gernskey, Lo, and Wei (2004). U.S. mutual fund asset totals are calculated using data from the CRSP Survivorship-Bias Free Mutual Fund Database (excluding money market funds and municipal bond funds). For hedge funds, "Debt" includes fixed income arbitrage and convertible arbitrage funds; "Domestic equity" includes long-short equity, equity market-neutral, event-driven, and short funds; "International" includes emerging markets and global macro funds; and "Other" includes managed futures, multi-strategy, and funds of funds. For mutual funds, "Debt" includes corporate and government bond funds (but neither money market funds nor municipal bond funds); "Domestic equity" includes all U.S. growth funds, balanced funds, and sector funds; and "International" includes all global and international equity and bond funds. At the bottom of each column, we report the Pearson correlation coefficient between the hedge fund and mutual fund data series.

Year	Share by broad category				
	TNA (\$billions)	Debt	Domestic equity	International	Other
Panel A: Global hedge fund industry					
1994	58	8%	32%	30%	31%
1995	70	9%	35%	27%	30%
1996	93	10%	36%	27%	27%
1997	138	11%	35%	28%	26%
1998	143	11%	42%	21%	26%
1999	175	10%	52%	14%	25%
2000	157	9%	48%	4%	19%
2001	246	13%	57%	5%	25%
2002	278	15%	61%	6%	28%
2003	390	15%	45%	10%	31%
2004	404	16%	46%	11%	27%
Panel B: U.S. mutual fund industry					
1994	1,124	25%	62%	13%	
1995	1,490	21%	67%	12%	
1996	1,866	18%	69%	13%	
1997	2,414	16%	72%	12%	
1998	2,899	15%	74%	11%	
1999	3,610	12%	75%	13%	
2000	3,473	12%	75%	13%	
2001	3,097	15%	73%	12%	
2002	2,646	20%	69%	11%	
2003	3,496	17%	71%	12%	
2004	4,097	15%	71%	14%	
Correlation (HF, MF)	0.77	-0.18	0.66	0.11	

Table 8. Hedge fund and mutual fund industry sizes by category.**Table A1**

The decline of sole-managed mutual funds and the rise of anonymously managed mutual funds, 1993–2004.

This table reports the percentage of mutual funds classified as reporting one manager name (sole-managed), reporting two or more manager names (co-managed), or reporting no manager names (anonymously managed). Panel B corresponds to the full sample of mutual funds in the CRSP Survivorship-Bias Free Mutual Fund Database (except money market funds). Panel A corresponds to the subset of these funds for which we possess the manager name variable from Morningstar. In both cases the unit of observation is the mutual fund rather than the mutual fund share class. The fractions in the first six columns reflect the actual Morningstar and CRSP manager name variables. The fractions in the last three columns are adjusted for time-series changes in the rules that Morningstar and CRSP use to classify a mutual fund's management structure. In 1993–1996, Morningstar identified any fund with more than two named managers as anonymously managed. Therefore, we use the distribution of transitions in management type between 1996 and 1997 to impute management type in 1993–1996. In 1993–1999, the CRSP manager name variable occasionally reports a firm name rather than a manager name. We use the distribution of transitions from firm names to sole-managed, co-managed, and anonymously managed to adjust the aggregate CRSP statistics.

Panel A: Management classification according to Morningstar									
As reported in Morningstar manager name variable						Adjusted for changing definitions			
1 Manager	2 Managers	3 Managers	4+ Managers	Anonymous	Firm Name	Sole-managed	Co-managed	Anonymous	
1993	71.0%	16.7%	0.0%	0.0%	12.2%	0.0%	71.0%	25.1%	3.9%
1994	69.7%	18.2%	0.0%	0.0%	12.2%	0.0%	69.7%	26.4%	3.9%
1995	67.9%	18.4%	0.0%	0.0%	13.7%	0.0%	67.9%	27.7%	4.4%
1996	62.5%	20.5%	0.0%	0.0%	17.0%	0.0%	62.5%	32.1%	5.4%
1997	57.8%	22.0%	8.2%	4.5%	7.6%	0.0%	57.8%	34.6%	7.6%
1998	52.3%	23.2%	8.8%	5.0%	10.7%	0.0%	52.3%	37.0%	10.7%
1999	49.8%	22.8%	9.2%	6.2%	12.0%	0.0%	49.8%	38.2%	12.0%
2000	47.2%	24.0%	9.2%	6.5%	13.1%	0.0%	47.2%	39.7%	13.1%
2001	45.4%	22.3%	9.6%	6.2%	16.4%	0.0%	45.4%	38.2%	16.4%
2002	43.6%	25.5%	10.9%	8.8%	11.1%	0.0%	43.6%	45.3%	11.1%
2003	41.6%	24.6%	9.8%	7.5%	16.5%	0.0%	41.6%	42.0%	16.5%
2004	40.6%	23.5%	9.7%	7.8%	18.3%	0.0%	40.6%	41.1%	18.3%
Panel B: Management classification according to CRSP									
As reported in CRSP manager name variable						Adjusted for changing definitions			
1 Manager	2 Managers	3 Managers	4+ Managers	Anonymous	Firm Name	Sole	Co-managed	Anonymous	
1993	74.5%	10.1%	1.8%	0.2%	5.2%	8.3%	79.2%	13.7%	7.1%
1994	73.2%	12.4%	2.5%	0.6%	5.7%	5.6%	76.4%	16.6%	7.0%
1995	71.3%	13.8%	3.2%	0.8%	5.7%	5.2%	74.3%	18.8%	6.9%
1996	65.8%	17.0%	5.7%	0.8%	6.8%	2.9%	68.0%	24.3%	7.7%
1997	60.0%	20.3%	7.0%	1.8%	7.6%	2.4%	60.0%	29.6%	8.1%
1998	58.3%	20.8%	6.8%	1.5%	10.5%	2.1%	59.5%	29.6%	11.0%
1999	53.2%	19.8%	7.9%	1.1%	17.1%	0.9%	53.7%	29.0%	17.3%
2000	49.3%	20.2%	7.2%	0.9%	22.4%	0.1%	49.3%	28.2%	22.4%
2001	46.3%	20.7%	6.9%	0.7%	25.4%	0.1%	46.3%	28.3%	25.4%
2002	42.9%	21.2%	6.1%	0.4%	29.2%	0.1%	43.0%	27.8%	29.2%
2003	40.4%	20.8%	6.4%	1.1%	31.2%	0.1%	40.5%	28.3%	31.2%
2004	39.0%	21.2%	6.5%	2.2%	31.0%	0.1%	39.1%	29.9%	31.0%

Appendix Table A1. Decline of sole-managed funds and rise of anonymous funds.

Table A2

Comparing Morningstar and CRSP classifications to mutual fund filings for random sample of domestic equity funds in 2002.

To determine whether Morningstar or CRSP provides the more accurate information on management structure, we hand-collected data on management structure for a sample of domestic equity funds in 2002. Funds were put into four bins based on whether CRSP or Morningstar classified the funds as being anonymously managed. For the funds chosen at random within each bin, data on the actual management structure were hand-collected from Prospectuses and Statements of Additional Information available on SEC's EDGAR database. In all but one case, we were able to locate manager names or a phrase like "The Adviser manages the Funds by an investment team approach" followed by no names. In the one case where we were not able to locate any explicit discussion of how the fund was managed, we followed both CRSP and Morningstar and classified the fund as anonymously managed. The "Number of domestic equity funds in 2002" is restricted to those mutual funds for which we possess the Morningstar manager name variable.

Morningstar classification	CRSP classification	Random sample of domestic equity funds	Number of filings that do not disclose manager names	Actual percentage anonymous	Number of domestic equity funds in 2002	Random sample as percentage population	Implied Morningstar accuracy	Implied CRSP accuracy
Named	Named	20	0	0.0%	1,316	1.5%	100.0%	100.0%
Named	Anonymous	45	8	17.8%	375	12.0%	82.2%	17.8%
Anonymous	Named	20	12	60.0%	42	47.6%	60.0%	40.0%
Anonymous	Anonymous	45	40	88.9%	135	33.3%	88.9%	88.9%
		130			1,868		94.7%	81.3%

Appendix Table A2. Comparing Morningstar and CRSP classifications to mutual fund filings.